Stoichiometry

Answer key with worked solutions

A. Mole ratios

1. Mole ratios in $4Al + 3O_2 \rightarrow 2Al_2O_3$.

> (a) $O_2 : Al$ = **3 mol : 4 mol** (b) $Al_2O_3 : Al$ = **2 mol : 4 mol = 1 : 2**

2. O_2 for 6.0 mol Al.

> $$6.0 \text{ mol Al} \times \frac{3 \text{ mol } O_2}{4 \text{ mol Al}} = \textbf{4.5 mol} \text{ of } O_2$$

B. Mass calculations

3. MgO from 12.15 g Mg.

> n(Mg) = 12.15 g ÷ 24.31 g/mol = 0.4998 mol → n(MgO) = 0.4998 mol (2 mol MgO : 2 mol Mg). M(MgO) = 24.31 g/mol + 16.00 g/mol = 40.31 g/mol, so mass = 0.4998 mol × 40.31 g/mol = **20.15 g**

4. O_2 from 24.5 g $KClO_3$.

> M($KClO_3$) = 122.55 g/mol; n = 24.5 g ÷ 122.55 g/mol = 0.1999 mol. n(O_2) = 0.1999 mol $KClO_3$ × (3 mol O_2 / 2 mol $KClO_3$) = 0.2999 mol. Mass = 0.2999 mol × 32.00 g/mol = **9.60 g**

5. CO_2 from 8.02 g CH_4.

> n(CH_4) = 8.02 g ÷ 16.04 g/mol = 0.500 mol → n(CO_2) = 0.500 mol (1 mol CO_2 : 1 mol CH_4). Mass = 0.500 mol × 44.01 g/mol = **22.0 g**

6. H_2O from 11.0 g C_3H_8.

> n(C_3H_8) = 11.0 g ÷ 44.09 g/mol = 0.2495 mol → n(H_2O) = 0.2495 mol C_3H_8 × (4 mol H_2O / 1 mol C_3H_8) = 0.998 mol. Mass = 0.998 mol × 18.02 g/mol = **18.0 g**

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C. Challenge

7. Balance, then find mass of Fe from 100.0 g Fe_2O_3.

> (a) $Fe_2O_3 + 3\,CO \longrightarrow 2\,Fe + 3\,CO_2$
>
> (b) n(Fe_2O_3) = 100.0 g ÷ 159.70 g/mol = 0.62617 mol → n(Fe) = 0.62617 mol Fe_2O_3 × (2 mol Fe / 1 mol Fe_2O_3) = 1.2523 mol. Mass = 1.2523 mol × 55.85 g/mol = **69.94 g**

8. The 2 g : 1 g mistake.

> Coefficients compare **moles**, not grams. 2 mol of Mg (2 mol × 24.31 g/mol = 48.62 g) reacts with 1 mol of O_2 (32.00 g); the mass ratio is not 2 : 1.

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